products, in a setup closer to the real scenario, and see how many years it takes you to find the best solution by hand!

We could already be happy with this solution, it’s the best we can find given the constraint.

1.4.4 Thinking (literally) out of the Box

But, in our narration, this is the point when the real expert in algorithms comes in. For instance, imagine that a distinguished academic is visiting our facilities while preparing the mission, invited to help with computing the best route to save fuel. During lunch break someone proudly tells her about how you brilliantly solved the problem with packing goods. And then she asks an apparently naïve question: why can’t you change the size of the boxes?

The answer will likely be either that “this is the way it has always been”, or that “items come already packed from vendors, and it would cost time and money to change it”.

And that’s when the expert will explain that if we remove the constrain on the box size, the 0-1 knapsack problem, which is an NP-complete problem, becomes the unbounded knapsack problem, for which we have a linear-time greedy solution that is usually better than the solution to the solution for the 0-1 version.

Translated in human-understandable language, we can turn this problem into one that’s easier to solve and that will